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$$\begin{aligned}\int_0^1 \ln(x) dx &= \lim_{a \rightarrow 0+} \int_a^1 \ln(x) dx \\&= \lim_{a \rightarrow 0+} [x \ln(x)]_a^1 - \lim_{a \rightarrow 0+} \int_a^1 1 dx \\&= \lim_{a \rightarrow 0+} (1 \ln 1 - a \ln a) - \lim_{a \rightarrow 0+} [x]_a^1 \\&= \left(0 - \lim_{a \rightarrow 0+} a \ln a\right) - (1 - 0)\end{aligned}$$

(onbepaalde vorm $0 \cdot \infty$, we herschrijven en passen L'Hôpital toe)

$$\begin{aligned}&= - \lim_{a \rightarrow 0+} \frac{\ln a}{\frac{1}{a}} - 1 \left(\frac{-\infty}{+\infty} \right) \\&= - \lim_{a \rightarrow 0+} \frac{\frac{1}{a}}{-\frac{1}{a^2}} - 1 \\&= - \lim_{a \rightarrow 0+} (-a) - 1 \\&= 0 - 1 \\&= -1\end{aligned}$$

References